# Hotel Booking Cancellation Analysis

## Data Analysis Project Report

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**Tools Used : Python, Pandas, Matplotlib, Seaborn**

**Dataset : Hotel Booking Demand Dataset (Kaggle)**

**Date : 15 March 2026**

# Introduction

Hotel booking cancellations are a major challenge in the hospitality industry as they can affect revenue and resource planning. This project analyzes the Hotel Booking Demand Dataset to understand patterns and factors related to booking cancellations. Using data analysis and visualization techniques, the dataset is explored to identify trends in customer behavior. Tools such as Python, Pandas, Matplotlib, and Seaborn are used for data cleaning and analysis. The goal of this project is to gain insights that can help hotels improve booking strategies and reduce cancellation rates.

# Business Problem

In Recent years, city Hotel and Resort Hotel have seen high Cancellations rates. Each hotel is now dealing with a number of issues as a result, including fewer revenues and less than ideal hotel room use. Consequently , lowering cancellation rates is both hotels primary goal in order to increase their efficiency in generating revenue, and for us to offer thorough business advice to address this problem.

The analysis of hotel booking cancellations as well as other factors that have no bearing on their business and yearly revenue generation are the main topics of this report.

Assumptions

1. No unusal occurrences between 2015 and 2017 will have a substantial impact on the data used.
2. The information is still current and can be used to analyze a hotel's possible plans in an efficient manner.
3. There are no unanticipated negatives to the hotel employing any advised technique.
4. The hotels are not currently using any of the suggested solutions.
5. The biggest factor affecting the effectiveness of earning income is booking cancellations.
6. Cancellations result in vacant rooms for the booked length of time.
7. Clients make hotel reservations the same year they make cancellations.

# Research Question

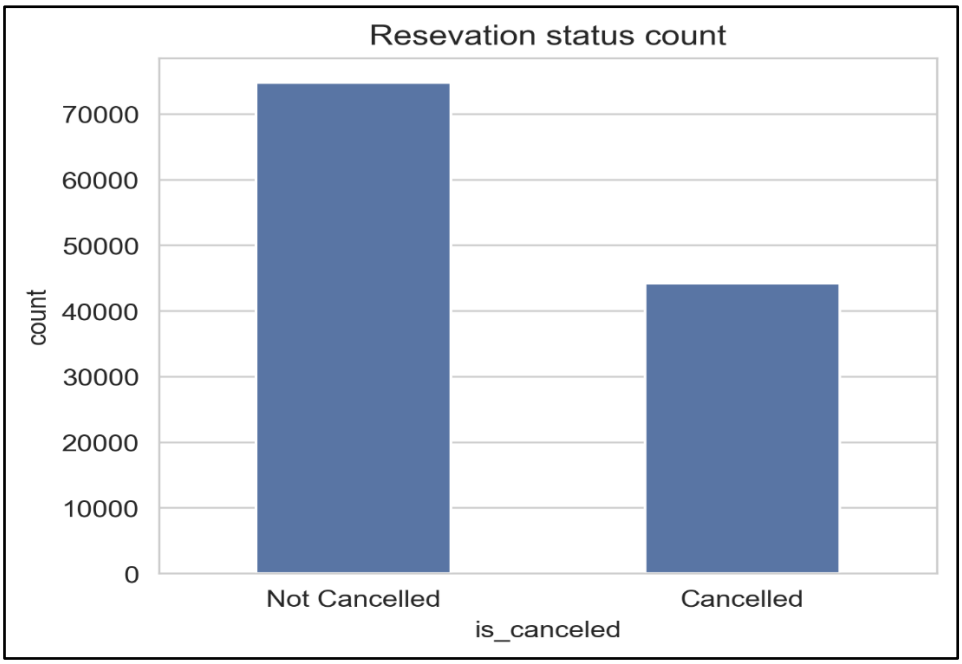
1. What are the variables that affect hotel reservation cancellations?
2. How can we make hotel reservations cancellations better?
3. How will hotels be assisted in making pricing and promotional decisions?

# Hypothesis

1. More cancellations occur when prices are higher.
2. When there is a longer waiting list, customers tend to cancel more frequently.
3. The majority of clients are coming from offline travel agents to make their reservations.

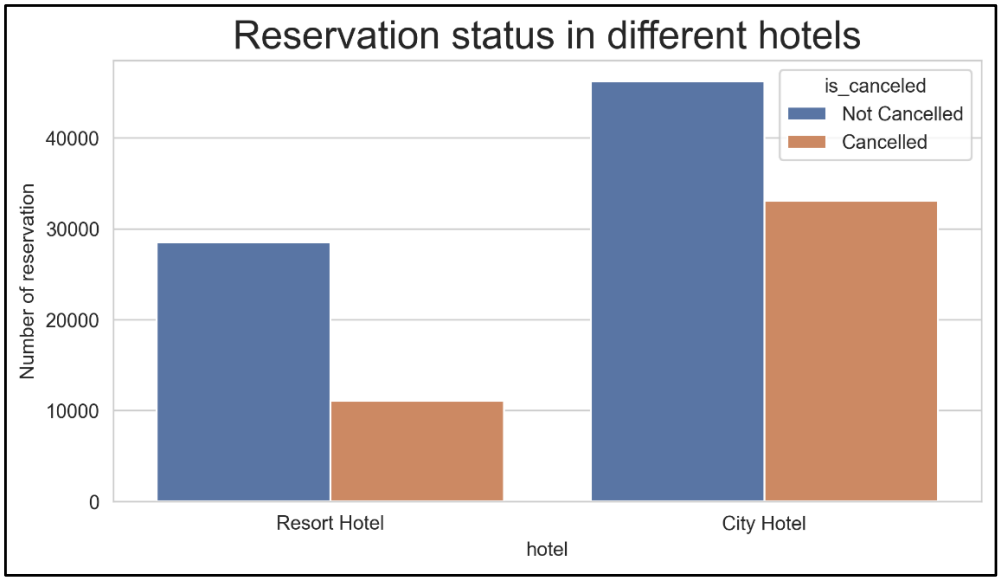
# Analysis and Findings

## 1. Count of Booking Cancellations



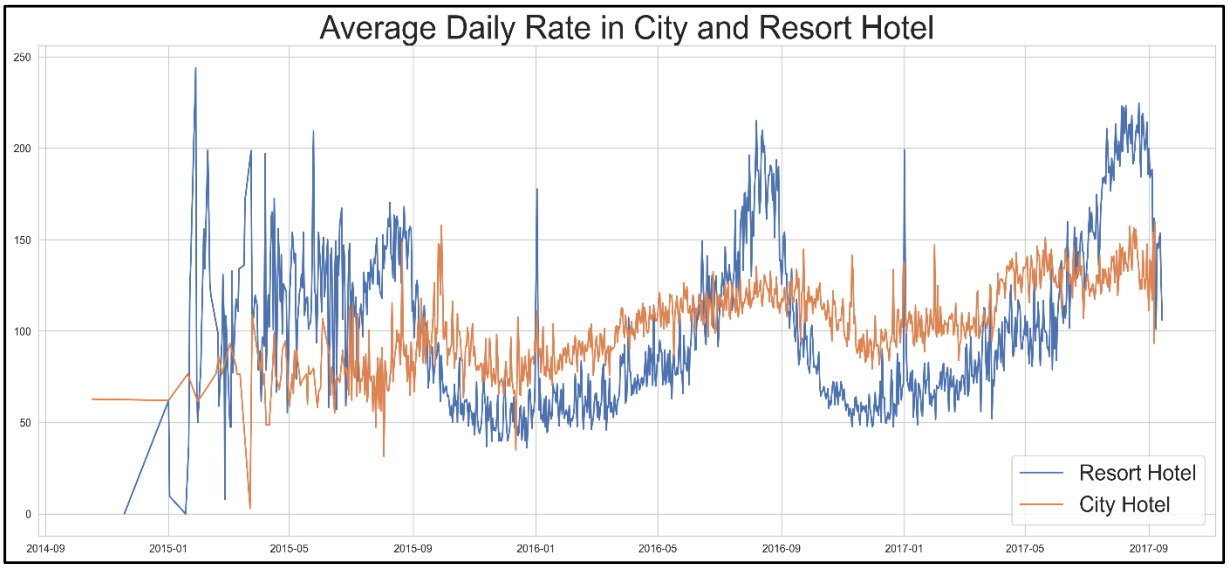

The Accompanying bar graph shows the percentage of reservations that are cancelled and those that are not. It is obvious that there are still a significant number of reservations that have been not cancelled. There are still 37% of clients who cancelled their reservation , which has a significant impact on the hotels earnings.

## 2. Reservation Status In Different hotels



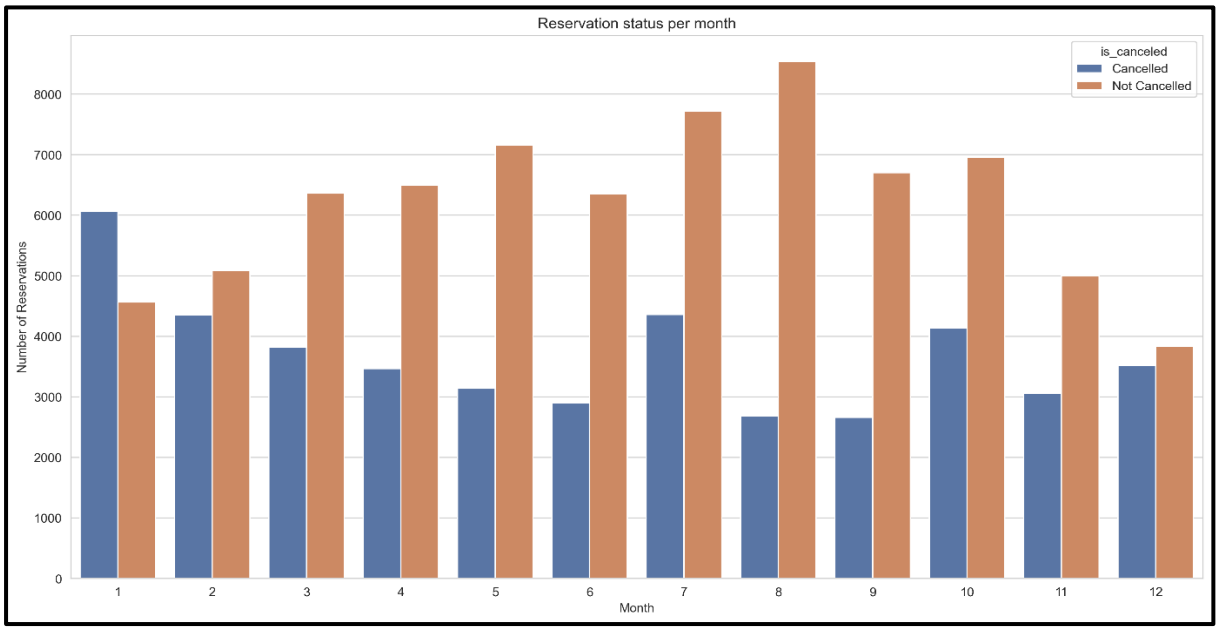

In Comparison to resort hotels, city hotels have more bookings. It’s possible that resort hotels are more expensive than those in cities.

## 3. Average Daily Rate In Hotels



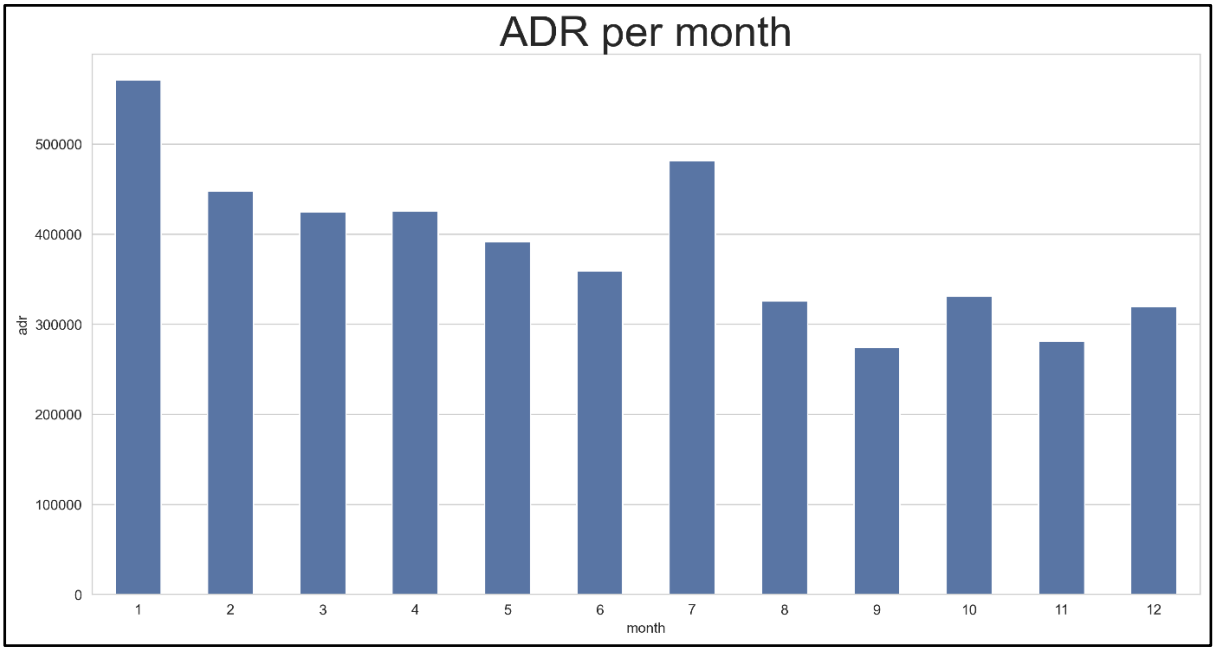

The line graph above shows that, on certain days, the average daily rate for a city hotel is less than that of a resort hotel, and on other days, it is even less. It goes without saying that weekends and holidays may see a rise in resort hotel rates.

## 4. Reservation Status Per Month



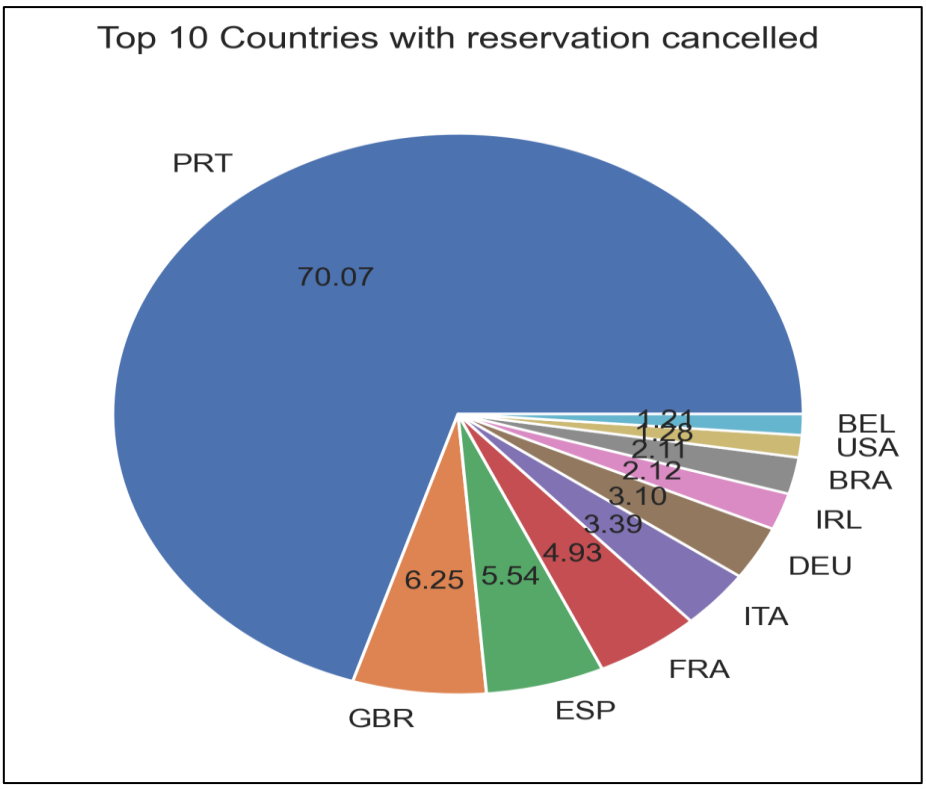
We have developed the grouped bar graph to analyze the months with the highest and lowest reservation levels according to reservation status. As can be seen, both the number of confirmed reservations and the number of cancelled reservations are largest in the month of August whereas January  is the month with the most cancelled reservations.

## 5. ADR For Each Month



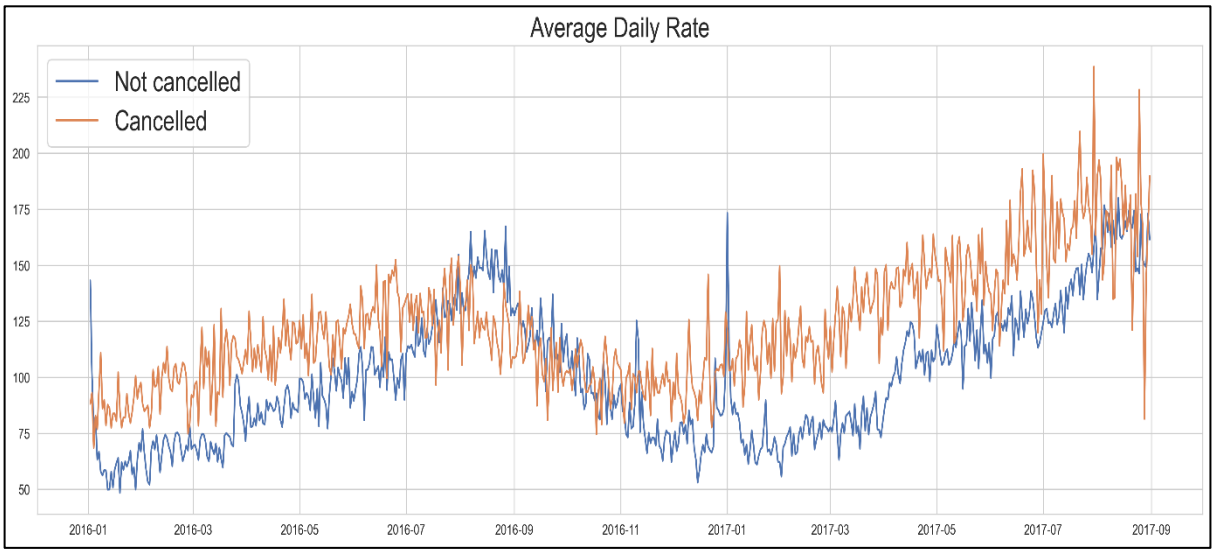

The bar graph demonstrates that cancellations are most common when prices are greatest and are least common when they are lowest. Therefore, the cost of the accommodation is solely responsible for the cancellation.

## 6. Top 10 Countries With Reservation Cancellations



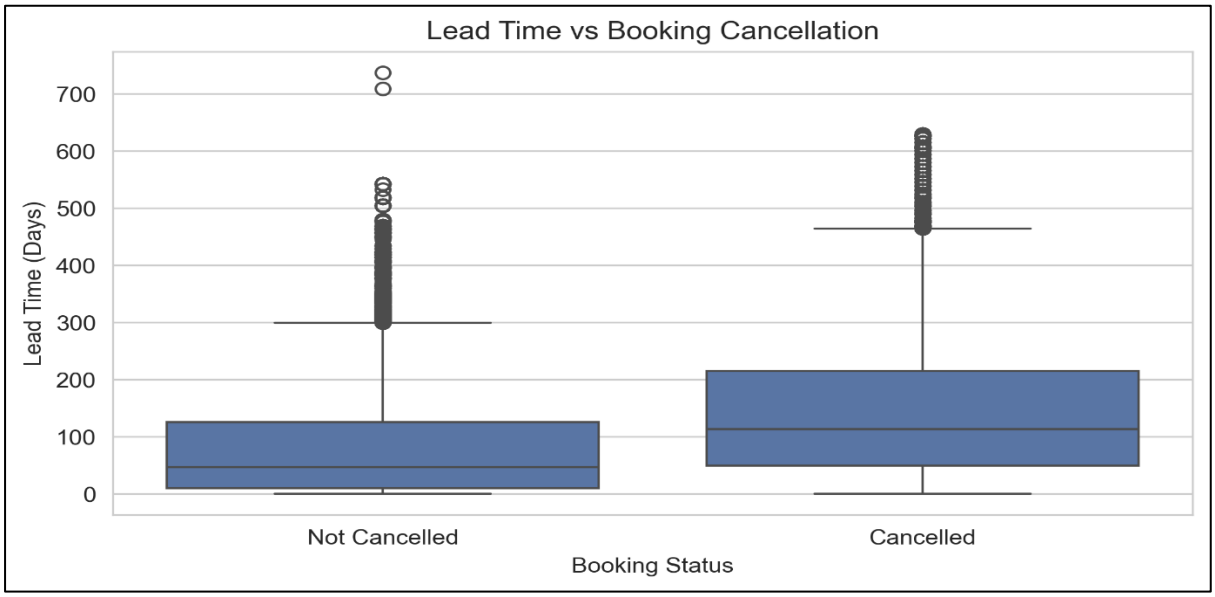

The pie chart shows the distribution of booking cancellations across different countries. It can be observed that Portugal has the highest number of cancellations compared to other countries. This indicates that a large portion of cancelled bookings comes from customers located in Portugal.

## 7. Average Daily Rate



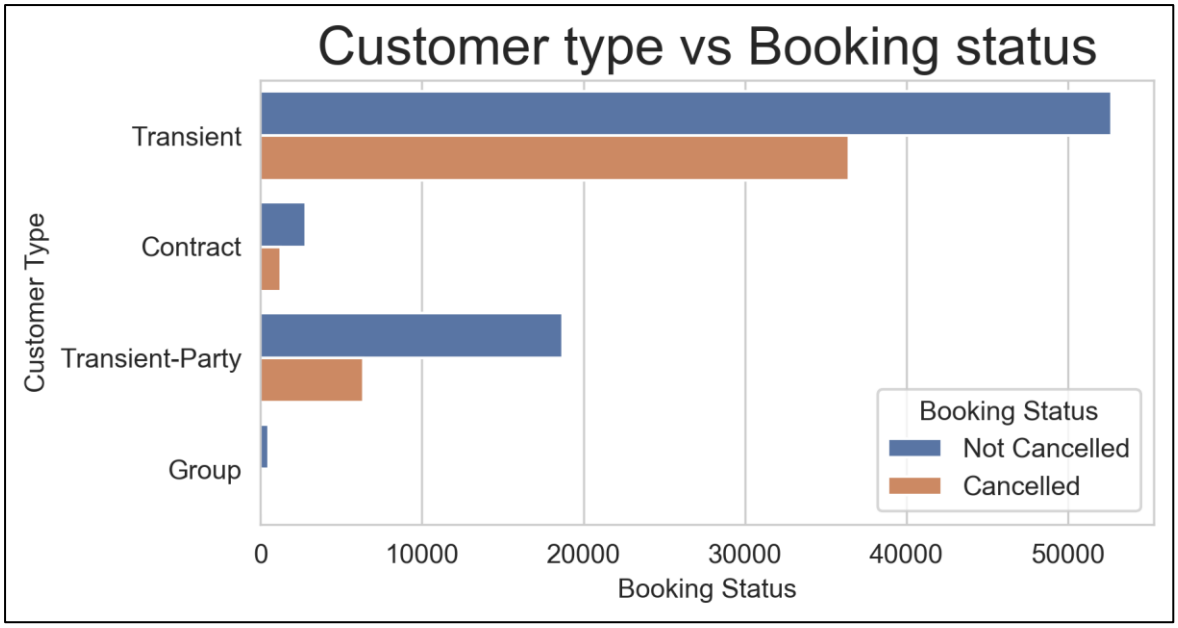
As seen in the graph , reservations are cancelled when the average daily rate is higher than when it in not cancelled. It clearly proves all the above analysis, that the higher price leads to higher cancellations.

## 8. Impact of Lead Time on Booking Cancellations




This chart analyses the relationship between lead time and booking cancellation. It can be observed that bookings with longer lead times are more likely to be cancelled compared to bookings made closer to the arrival date. This suggests that customers who book much earlier tend to change their plans more frequently.

## 9. Customer Type Distribution in Booking Cancellations




This chart shows that cancellation pattern among different customer types. It can be observed that certain customer types have a higher cancellation rate than others. This insight helps hotels understand which customer segment is more likely to cancel their bookings.

# Suggestions

1. Cancellation rates rise as the price does. In order to prevent cancellations of reservations, hotels could work on their pricing strategies and try to lower the rates for specific hotels based on locations. They can also provide some discounts to the consumers.
2. As the ratio of the cancellations and not cancelled of the resort hotels is higher in the resort hotel than the city hotels. So the hotels should provide a reasonable discount on the room prices on weekends or on holidays.
3. In the month of January, hotels can start campaigns or marketing with a reasonable amount to increase their revenue as the cancellation is the highest in this month.
4. They can also increase the quality of their hotels and their services mainly in Portugal to reduce the cancellation rate.